

DANIEL BAIG

d.p.baig@soton.ac.uk • <https://github.com/danielbaig>

PROFILE

Pursuing doctoral studies in topics related to gauge/gravity duality at the University of Southampton.
Possesses 8 months of cumulative research experience developed through a number of research internships.
Solves problems by applying mathematical and coding experience to study a variety of physical systems.
Interested in pursuing a career in theoretical physics research.

RESEARCH EXPERIENCE

UNIVERSITY OF CAMBRIDGE, Cambridge, UK

Jul - Aug 2025

Sainsbury Laboratory Cambridge University (SLCU)

Cambridge Mathematics Placement

- Completed an 8-week project on modelling self-similar plant structures using a custom numerical solver for morphodynamics, namely Organism.
- Created models for the Shoot Apical Meristem and apical hook that replicated growth seen in confocal images.

UNIVERSITY OF OXFORD, Oxford, UK

Jul - Sep 2023

Department of Physics

Astrophysics Summer Research Programme

- Undertook an 8-week project on modelling disk winds from the accretion disk of X-ray binaries using the Monte Carlo radiative transfer code: PYTHON.
- Devised a method for approximating the movement of points in the output parameter space using a Monte Carlo string energy minimisation method.
- Successfully reproduced P-Cygni blueshifted absorption lines in the optical spectrum for an X-ray binary as seen in observations for the first time.

UNIVERSITY COLLEGE LONDON, London, UK

May - Jul 2023

UCL High Energy Physics (HEP)

MAPS Research Internship

- Undertook a 6-week project on analysing data from the LHC using the CONTUR package in Python to constrain the parameter space of dark meson masses in a composite dark matter model.
- Created code to verify simulated cross-sections from RIVET with detector data.
- Results of the project are available at:
<<https://hepcedar.gitlab.io/contur-webpage/results/HeavyDarkMesons/index.html>>

THE FRANCIS CRICK INSTITUTE, London, UK

May - Jul 2022

UCL Institute for the Physics of Living Systems (IPLS)

Biophysics Research Internship

- Completed a 10-week project on using coupled, non-linear, differential equations to simulate the evolution of a gene regulatory network in Python to explore convergence in evolution.
- Delivered two presentations to other IPLS students and their supervisors on the aims and results of the project. Additionally, wrote a report on the outcomes of the investigation.

EDUCATION

UNIVERSITY OF SOUTHAMPTON, Southampton, UK

Sep 2025 - Present

PhD Mathematical Sciences

- *Research Focus* - Gauge/Gravity Duality
- *Supervisors* - Dr Fridrik Gautason and Dr Benjamin Withers.

UNIVERSITY OF CAMBRIDGE, Cambridge, UK

Oct 2024 - Jun 2025

MASt Theoretical Physics (Part III of the Mathematical Tripos) - Merit

- *Essay Title* - Positivity of Bondi Energy in General Relativity.
- *Relevant Modules* - Quantum Field Theory, General Relativity, Advanced Quantum Field Theory, String Theory, and Black Holes.
- *Audited Modules* - Gauge/Gravity Duality; Solitons, Instantons and Geometry; Canonical Gravity; Cosmology; Symmetries, Particles and Fields; and Statistical Field Theory.

UNIVERSITY COLLEGE LONDON, London, UK

Sep 2020 - Jun 2024

MSci Theoretical Physics - First Class Honours

- *Dean's list* - Top 5% of the Faculty of Mathematical and Physical Sciences.
- *Winton Prize* - Best overall academic achievement by a UCL 4th year MSci Theoretical Physics student.
- *Final Year Project* - Uncertainties on Parton Distributions.
- *Relevant Modules* - Quantum Field Theory, Mathematics for General Relativity, Practical Machine Learning for Physicists, Techniques of High-Performance Computing and Cosmology.

SKILLS AND INTERESTS

- Adept in **Python**, including NumPy, pandas, Astropy, Numba, SciPy and Keras packages.
- Ability to model physical systems in **Mathematica** and **C++**, utilising Eigen, gnuplot and SDL libraries.
- Proficient in producing reports and documents using LaTeX/Overleaf.
- Experienced in using the Unix shell in Linux to run simulations written in Python and C on a cluster developed in the research experiences listed above.
- Constructs programs in free time to develop coding aptitude and learn how to apply new concepts.

REFERENCES

Referee	Relation	Email
Prof Andrew Pontzen	UCL Personal Tutor	andrew.p.pontzen@durham.ac.uk
Prof Robert Thorne	UCL Final Project Supervisor	robert.thorne@ucl.ac.uk
Dr James Matthews	Oxford Astrophysics Supervisor	james.matthews@physics.ox.ac.uk
Prof Jonathan Butterworth	UCL HEP Supervisor	j.butterworth@ucl.ac.uk